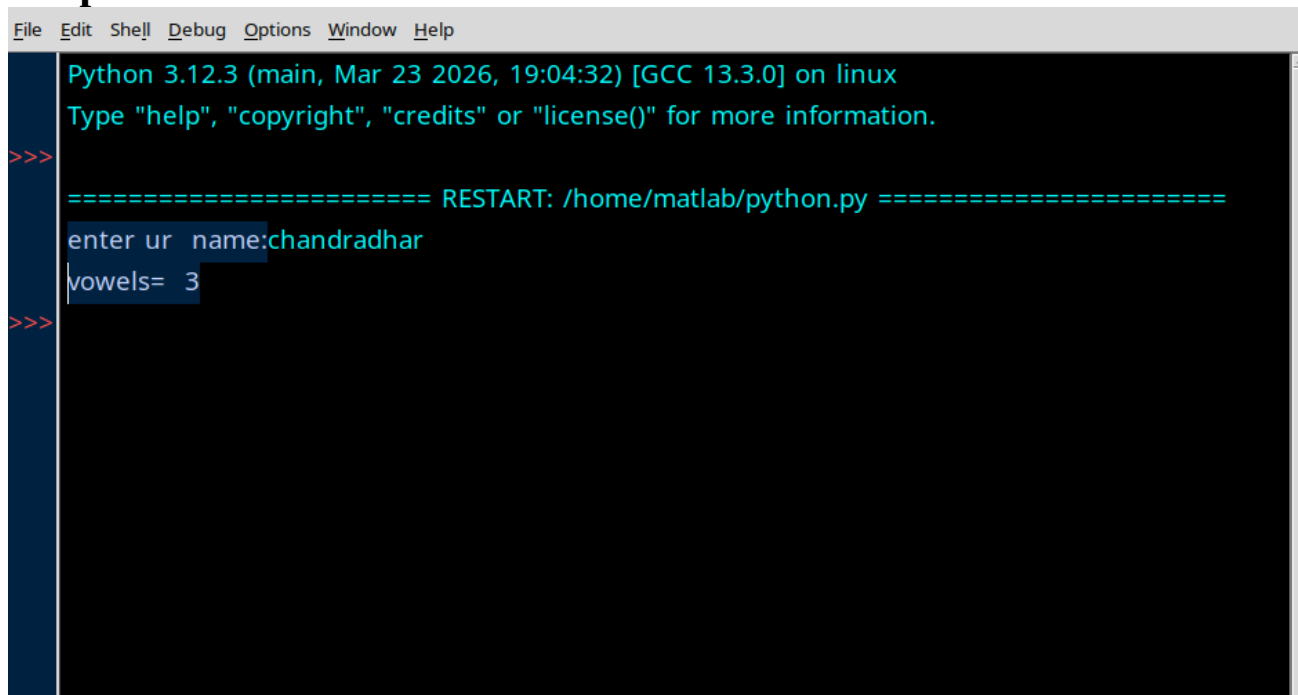


Q1) Count number of vowels using for loop

Code:

```
a=str(input("enter ur name:"))
c=0
for i in a:
    if i=='a' or i=='e' or i=='i' or i=='o' or i=='u':
        c+=1
print("vowels= ", c)
```

Output:



The screenshot shows a terminal window with a menu bar (File, Edit, Shell, Debug, Options, Window, Help). The terminal text is as follows:

```
Python 3.12.3 (main, Mar 23 2026, 19:04:32) [GCC 13.3.0] on linux
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: /home/matlab/python.py =====
enter ur name:chandradhar
vowels= 3
>>>
```

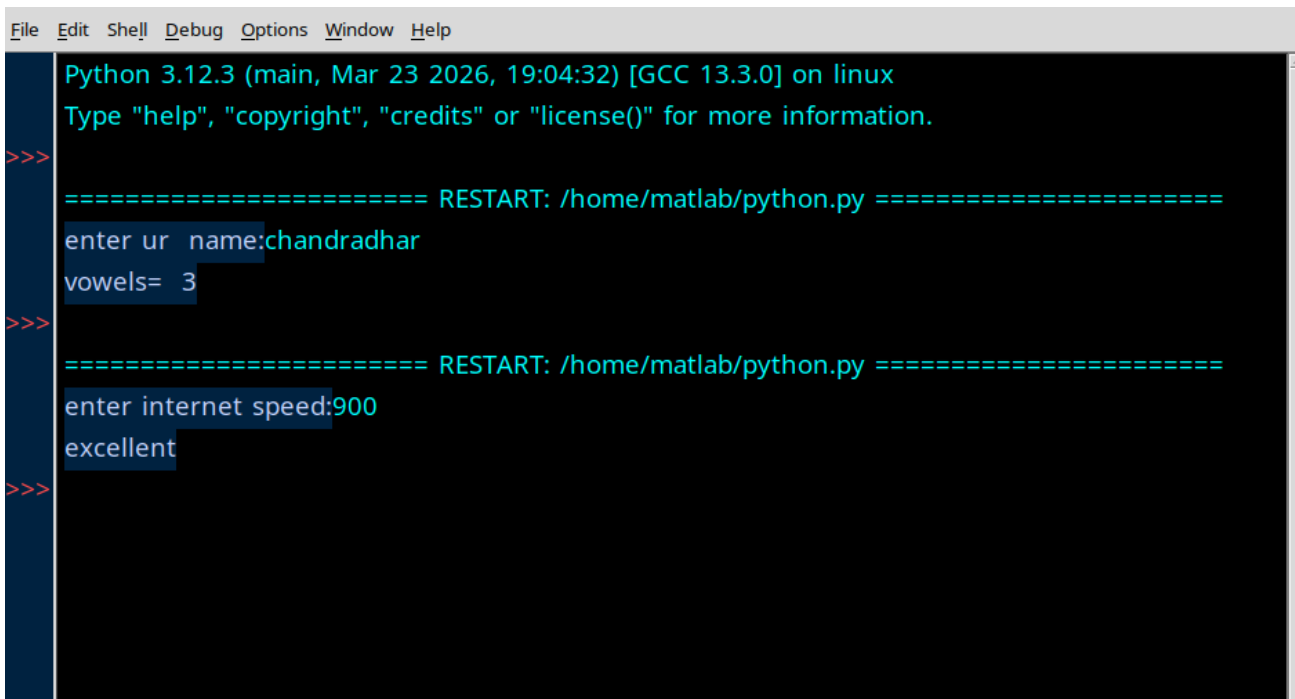
Q2) Calculating Internet speed rating using if-else conditions

Code:

```
s=int(input("enter internet speed:"))
if s>=100:
    print("excellent")
elif s>=50:
```

```
    print("good speed")
elif s>=20:
    print("average speed")
else:
    print("Worst")
```

Output:



The screenshot shows a Python 3.12.3 terminal window. The prompt is >>>. The user enters 'enter ur name:chandradhar' and 'vowels= 3'. The program restarts. The user enters 'enter internet speed:900' and 'excellent'. The program restarts again.

```
Python 3.12.3 (main, Mar 23 2026, 19:04:32) [GCC 13.3.0] on linux
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: /home/matlab/python.py =====
enter ur name:chandradhar
vowels= 3
>>>
===== RESTART: /home/matlab/python.py =====
enter internet speed:900
excellent
>>>
```

Q3) Countdown timer using loops

Code:

```
n=int(input("enter countdown"))
for i in range(n,0,-1):
    print(i)
print("time's up!!!")
```

Output:

```
File Edit Shell Debug Options Window Help
Python 3.12.3 (main, Mar 23 2026, 19:04:32) [GCC 13.3.0] on linux
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: /home/matlab/python.py =====
enter ur name:chandradhar
vowels= 3
>>>
===== RESTART: /home/matlab/python.py =====
enter internet speed:900
excellent
>>>
===== RESTART: /home/matlab/python.py =====
enter countdown10
10
9
8
7
6
5
4
3
2
1
time's up!!!
>>>
```

Q4) Finding a largest word using lists

Code:

```
w= input("Enter words: ").split()
longest= max(w, key=len)
print("Longest word:", longest)
```

Output:

```
Output Clear
Enter words: apple ball cat dolphin
Longest word: dolphin

=== Code Execution Successful ===
```

Q5) Unique letter in a word using sets

Code:

```
w = input("Enter a word: ")
l = set(w)
print("Unique letters:", l)
```

```
print("Count:", len(l))
```

Output:

```
Output Clear
Enter a word: Antibacterial
Unique letters: {'r', 'n', 'c', 'e', 'i', 'A', 'l', 't', 'a', 'b'}
Count: 10

=== Code Execution Successful ===
```

Q6) Calculating parking fee using functions:

Code:

```
def pf(hours):
    return hours * 20
h = int(input("Enter parking hours: "))
print("Fee =", pf(h))
```

Output:

```
Output Clear
Enter parking hours: 12
Fee = 240

=== Code Execution Successful ===
```

Q7) Fruit Price using Dictionaries

Code:

```
prices = {
    "apple": 120,
    "banana": 40,
    "kiwi": 80,
    "mango": 150
}
fruit = input("Enter fruit name: ").lower()
if fruit in prices:
```

```
    print("Price =", prices[fruit])
else:
    print("Fruit not available")
```

Output:

```
Output Clear
Enter fruit name: apple
Price = 120

=== Code Execution Successful ===
```

Q8) Star pattern Printing using nested loops:

Code:

```
rows = int(input("Enter number of rows: "))
for i in range(1, rows + 1):
    print("*" * i)
```

Output:

```
Output Clear
Enter number of rows: 4
*
**
***
****

=== Code Execution Successful ===
```

Q9) Palindrome Checker using strings

Code:

```
text = input("Enter a word: ")
r = ""
for ch in text:
```

```
r = ch + r
if text == r:
    print("Palindrome")
else:
    print("Not a palindrome")
```

Output:

Output Clear

```
Enter a word: markram
Palindrome

=== Code Execution Successful ===
```

Q10) Counting Number Frequency using Dictionaries

Code:

```
numbers = list(map(int, input("Enter numbers with space: ").split()))
f= {}
for i in numbers:
    if i in f:
        f[i] += 1
    else:
        f[i] = 1
print("Number Frequency:")
for key, value in f.items():
    print(key, "->", value)
```

Output:

Output Clear

```
Enter numbers with space: 1 3 2 4 3 2 2 3 4 2 3 4
Number Frequency:
1 -> 1
3 -> 4
2 -> 4
4 -> 3

=== Code Execution Successful ===
```

